Fall 2025 CPTS530 Homework 03 Report
Numerical Analysis

Aryan Ritwajeet Jha

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Question 1

Problem Statement: Suppose that $x > 0$, $y < 0$. Using definition of absolute value and splitting into cases, show that $|x + y| \leq |x - y|$.

Solution:
Given: $x > 0$, $y < 0$
To show: $|x + y| \leq |x - y|$

Visual Representation:

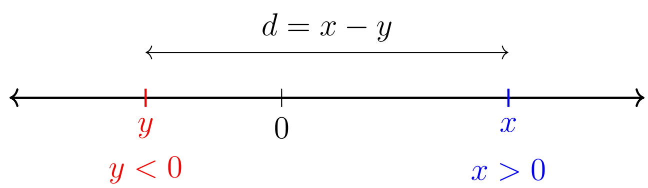

Figure 1: Number line showing the relationship between x, y and $x - y$

Since $x > 0$ and $y < 0$, we can assume that $x - y = d$ where $d > 0$.
Therefore, RHS $= |x - y| = d$ is always true.
Now we need to consider cases for $|x + y|$ based on whether $x + y$ is positive or negative.

Case I: $x + y \geq 0$
From $x - y = d$, we get $x = y + d$.
Therefore: $x + y = (y + d) + y = 2y + d$
Since $x + y \geq 0$, we have $|x + y| = x + y = 2y + d$.
So LHS $= 2y + d$ and RHS $= d$.
Since $y < 0$, we have $2y < 0$, which means LHS $= 2y + d < d =$ RHS.

Case II: $x + y < 0$
From $x - y = d$, we get $y = x - d$.
Therefore: $x + y = x + (x - d) = 2x - d < 0$
Since $x + y < 0$, we have $|x + y| = -(x + y) = -(2x - d) = d - 2x$.
So LHS $= d - 2x$ and RHS $= d$.
Since $x > 0$, we have $2x > 0$, which means LHS $= d - 2x < d =$ RHS.

Conclusion:
Cases I and II exhaustively cover all possible scenarios. In both cases, we have shown that LHS < RHS, which means $|x + y| < |x - y|$.
Therefore, we have proven the stronger result: $\boxed{|x + y| \leq |x - y| \text{ for } x > 0, y < 0}$
Hence Proved. $\square$

Question 2

Problem Statement: Suppose you have an iterative algorithm for finding a fixed point of a function $F(x)$ on a certain interval. Write a short text (2-4 paragraphs) explaining the following:

1. How do you decide when to stop iterating?
2. What is your strategy for making sure that the computed solution is close to the exact solution?

DO NOT use the standard methodology and estimates from the textbook. Try to find other ways of accomplishing the goals above. Discuss briefly the advantages and disadvantages of your error control approach.

Solution:

Assumptions:

1. $F(x)$ is a contractive function in the given interval (I have the expression and can use it to generate x_n as $n = 1, 2, 3, \ldots$ and some x_0).
2. I start with an x_0 'close enough' to 'r', so that I can assume that $F(x)$ will converge to r with enough iterations, either because it was in its 'region of contraction' or even otherwise. Else by running my solvers/algorithms repeatedly I can switch x_0 till I get a reliable convergence to some fixed point r.

Approach:

From knowledge of $F(x)$, I can determine the order of convergence via Taylor's formula.

Step 1: Determine Order of Convergence from Table

Since I don't know the exact fixed point r, I cannot directly compute the errors $e_n = |x_n - r|$. However, I can estimate the order of convergence p using consecutive iterations.

Step 2: Table Generation

Generate a table of n vs x_n values by recursively using $x_{n+1} = F(x_n)$:

n	x_n
0	x_0
1	x_1
2	x_2
$\vdots$	$\vdots$
n	x_n
$\vdots$	$\vdots$

Step 3: Estimating the Order of Convergence p

For a convergent sequence with order of convergence p, I have:

$$\lim_{n\to\infty} \frac{|x_{n+1} - r|}{|x_n - r|^p} = C$$

where C is a constant and r is the unknown fixed point.

Since I don't know r, I can use the approximation that for some sufficiently large n, $x_n \approx r$. This gives me:

$$\frac{|x_{n+2} - x_{n+1}|}{|x_{n+1} - x_n|^p} \approx C$$

Taking logarithms of consecutive ratios:

$$\ln\left(\frac{|x_{n+2} - x_{n+1}|}{|x_{n+1} - x_n|}\right) \approx p \ln\left(\frac{|x_{n+1} - x_n|}{|x_n - x_{n-1}|}\right)$$

Rearranging to solve for p:

$$p \approx \frac{\ln\left(\frac{|x_{n+2}-x_{n+1}|}{|x_{n+1}-x_n|}\right)}{\ln\left(\frac{|x_{n+1}-x_n|}{|x_n-x_{n-1}|}\right)} \tag{1}$$

This formula allows me to estimate p using only the computed values x_{n-1}, x_n, x_{n+1}, and x_{n+2} without knowing the exact fixed point r.

When do I know that the computed solution is close enough to the exact solution?
Once the RHS of Equation 1 stabilizes to a constant value (i.e., the estimated p does not change significantly with further iterations), I can be confident that I'm 'heading towards' the fixed point r i.e. my FPI iterations are in the 'region of contraction'.

How do you decide when to stop iterating?
Once 'sufficient closeness' to r is determined, I can employ a tolerance on the absolute (approximate) error value, say ϵ,. I can then stop the iterations when my approximate e_n falls below this threshold and conclude that my approximate $x_n \approx r$. $\square$

Question 3

Problem Statement: Do Problem 23 c) from Section 3.4.

Problem 3.4.23 c): Find the order of convergence of the sequence:

$$x_n = \left(1 + \frac{1}{n}\right)^{\frac{1}{2}}$$

Solution:

Question 4

Problem Statement: Do Problem 32 from Section 3.4.

Problem 3.4.32: Let $\frac{1}{2} \le q \le 1$, and define $F(x) = 2x - qx^2$. On what interval can it be guaranteed that the method of iteration using F will converge to a fixed point? (This problem is related to Problem 3.2.5, p. 90.)

Related Problem 3.2.5: What is the purpose of the following iteration formula?

$$x_{n+1} = 2x_n - x_n^2 y$$

Identify it as the Newton iteration for a certain function.

Solution:

Analysis of Related Problem 3.2.5

First, let's identify what the iteration formula $x_{n+1} = 2x_n - x_n^2 y$ represents as a Newton iteration.

The Newton-Raphson method for solving $f(x) = 0$ has the form:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Comparing with our given iteration:

$$x_{n+1} = 2x_n - yx_n^2 = x_n - (-x_n + yx_n^2)$$

Therefore:

$$\frac{f(x_n)}{f'(x_n)} = yx_n^2 - x_n$$

To solve for $f(x_n)$ we'll have to use integration. Let's temporarily switch to x as a variable instead of x_n.

$$\frac{f(x)}{f'(x)} = yx^2 - x$$

or,

$$\frac{f(x)}{\frac{df(x)}{dx}} = yx^2 - x$$

Cross-multiplying with $\frac{1}{dx}$ on both sides:

$$\frac{f(x)}{df(x)} = \frac{yx^2 - x}{dx}$$

Equivalently, we can write:

$$\frac{df(x)}{f(x)} = \frac{dx}{yx^2 - x}$$

Integrating both sides:

$$\ln|f(x)| = \int \frac{dx}{yx^2 - x} = \int \frac{dx}{x(yx-1)}$$

Dividing numerator and denominator or RHS by y:

$$\ln|f(x)| = \int \frac{1/y}{x(x-1/y)} dx$$

Therefore:

$$\ln|f(x)| = \int \left(\frac{1}{x-1/y} - \frac{1}{x} \right) dx$$

$$\text{or,}\ \ln|f(x)| = \ln\left| x - \frac{1}{y} \right| - \ln|x| + \ln k\ ,\ k \neq 0$$

$$\text{or,}\ \ln|f(x)| = \ln\left| \frac{k(x-\frac{1}{y})}{x} \right|$$

Taking the exponential:

$$f(x) = \frac{k}{x}\left(x - \frac{1}{y} \right)$$

Giving us the root of $f(x)$ or alternatively the fixed point of $F(x)$ (let's call it r) at

$$x = r = \frac{1}{y}$$

.

Main Problem Solution

Now for the main problem with $F(x) = 2x - qx^2$ where $\frac{1}{2} \leq q \leq 1$.

Basically here $y = q$, so the fixed point is $r = \frac{1}{q}$.

For the fixed point iteration to converge, we need $|F'(x)| < 1$ in an interval $[a, b]$ containing the fixed point $r = \frac{1}{q}$.

Step 1: Find the derivative

$$F'(x) = 2 - 2qx$$

Step 2: Applying the convergence condition (from a previous theorem in homework 2) For convergence: $|F'(x)| < 1$

This gives us: $|2 - 2qx| < 1$

Step 3: Solve the inequality

$$-1 < 2 - 2qx < 1$$

For upper limit: $2 - 2qx > -1$

$$2qx < 3$$

$$x < \frac{3}{2q}$$

From lower limit: $2 - 2qx < 1$

$$2qx > 1$$

$$x > \frac{1}{2q}$$

Thus: $\frac{1}{2q} < x < \frac{3}{2q}$. We may note that the fixed point $r = \frac{1}{q}$ lies in this interval as well.

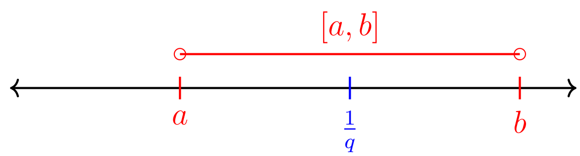

Figure 2: Convergence interval showing $a = \frac{1}{2q}$, fixed point $r = \frac{1}{q}$, and $b = \frac{3}{2q}$. The closed interval containing the extreme points is proven to be valid for convergence (see below).

Normally, a contractive function requires a closed interval like $[a, b]$, but here we have an open interval (a, b).

Since $F'(x) = 2 - 2qx$ is monotonically decreasing (as $q > 0$), we can verify that $F'(a) = 2 - 2q \cdot \frac{1}{2q} \leq 1$ and $F'(b) = 2 - 2q \cdot \frac{3}{2q} \geq -1$. This ensures that $|F'(x)| < 1$ throughout the open interval (a, b).

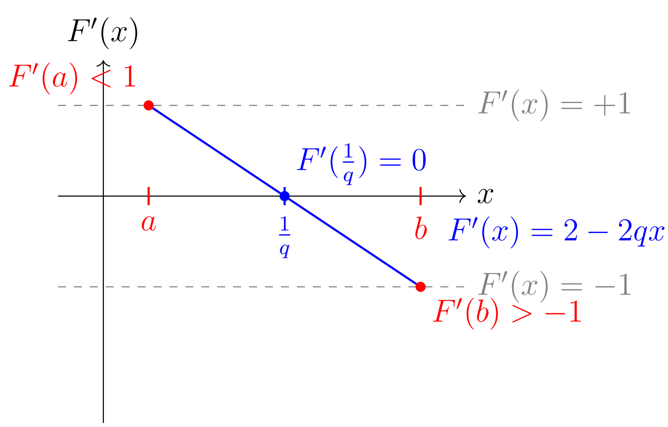

Figure 3: Monotonic (decreasing) behavior of $F'(x) = 2 - 2qx$ showing convergence condition $|F'(x)| < 1$ satisfied in (a, b) given that $q > 0$.

This does not mean that the extreme points a and b are necessarily ruled out. We can check for $F(a)$ and $F(b)$ to see if they 'get' mapped into the contractive region and if they both are, we can safely claim that the closed interval $[a, b]$ is also valid.

Calculating $F(a)$:

$$F(a) = F\left(\frac{1}{2q}\right) = 2\left(\frac{1}{2q}\right) - q\left(\frac{1}{2q}\right)^2 = \frac{1}{q} - \frac{1}{4q} = \frac{3}{4q}$$

Calculating $F(b)$:

$$F(b) = F\left(\frac{3}{2q}\right) = 2\left(\frac{3}{2q}\right) - q\left(\frac{3}{2q}\right)^2 = \frac{3}{q} - \frac{9}{4q} = \frac{3}{4q}$$

Thus we can see that both $F(a)$ and $F(b)$ map to $\frac{3}{4q}$ which lies in the interval (a, b).

Therefore we can safely say that the closed interval $[a, b] = \left[\frac{1}{2q}, \frac{3}{2q}\right]$ is also valid.

Now as q itself is given as a parameter in the range $\frac{1}{2} \leq q \leq 1$, we can see that the interval $\left[\frac{1}{2q}, \frac{3}{2q}\right]$ will vary based on the value of q.

We want to calculate the largest 'common' interval of [a,b] valid for all q in the given range.

Actually this answer depends on whether we have an apriori knowledge of q or not.

If we do know the value of q beforehand, then we can use the interval $\left[\frac{1}{2q}, \frac{3}{2q}\right]$ for that specific q.

However if q is strictly to be treated as a parameter between $\frac{1}{2}$ and 1, we're interested in some 'universal' interval where $F(x)$ will converge to a fixed point regardless of the value of q – also it may be noted that the fixed point itself is allowed to lie outside $[a, b]$.

We recognize that the values for a and b are monotonically dependent on the value of q, both a and b decrease as q increases.

We can then exploit this monotonicity to come up with a formula for the universal interval of convergence.

Step 2: Intersection of all these intervals

The intersection of all I_q over $q \in [0.5, 1]$ is:

$$\bigcap_{q \in [0.5,1]} I_q = [\max a(q), \min b(q)] = [a(0.5), b(1)] = \left[1, \frac{3}{2}\right]$$

Thus the largest common interval for all q in the range $\frac{1}{2} \leq q \leq 1$ is $\left[1, \frac{3}{2}\right]$.

Final Answer:

The method of iteration using $F(x) = 2x - qx^2$ will converge to the fixed point $r = \frac{1}{q}$ on the interval:

If q is known apriori:

$$\boxed{\left[\frac{1}{2q}, \frac{3}{2q}\right]}$$

If q is strictly a parameter in the range $\frac{1}{2} \leq q \leq 1$:

$$\boxed{\left[1, \frac{3}{2}\right]} \quad \square$$